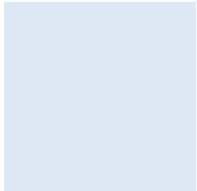
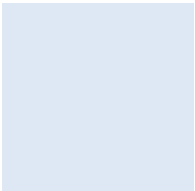
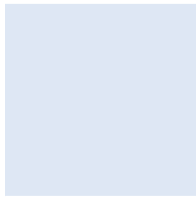


Info type: Confidential

Company: NTT Data Payment Services India

Info. owner: Product (PMG)



**Transaction API**

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## Document Information

| Document Attributes | Information                      |
|---------------------|----------------------------------|
| ID                  | Transaction API (Non-seamless)   |
| Owner               | Product Management & Engineering |
| Author              | P&E – Product Team               |

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# I. Payment Initiation API

## 1. Introduction

Transaction API (Non-Seamless) facilitates the payment process from the initiation of the transaction to the authorization of the payment by the customer on the Payment Page. The transaction-related data is exchanged **server-to-server** directly via **RESTful** APIs. This API acts as a method to obtain the payment pop-up post request authorization and get it displayed on the checkout page.

By integrating this API into your system, you can offer your customers a wide range of payment options. It is optimized with device screen size and helps customers to pay easily and securely over web or on mobile. It supports payment options such as Credit Cards, Debit Cards, Net Banking, UPI, Wallets & NEFT/RTGS.

It also provides a retry option for reducing customer drop-offs in case of transaction failure. Customer can retry the transaction by selecting alternate payment options.

## 2. Key Features

- Easy integration
- Optimized UI
- PCI-DSS complaint
- User-friendly Customer UI
- Provides retry option, reducing customer drop

## 3. Payment Process

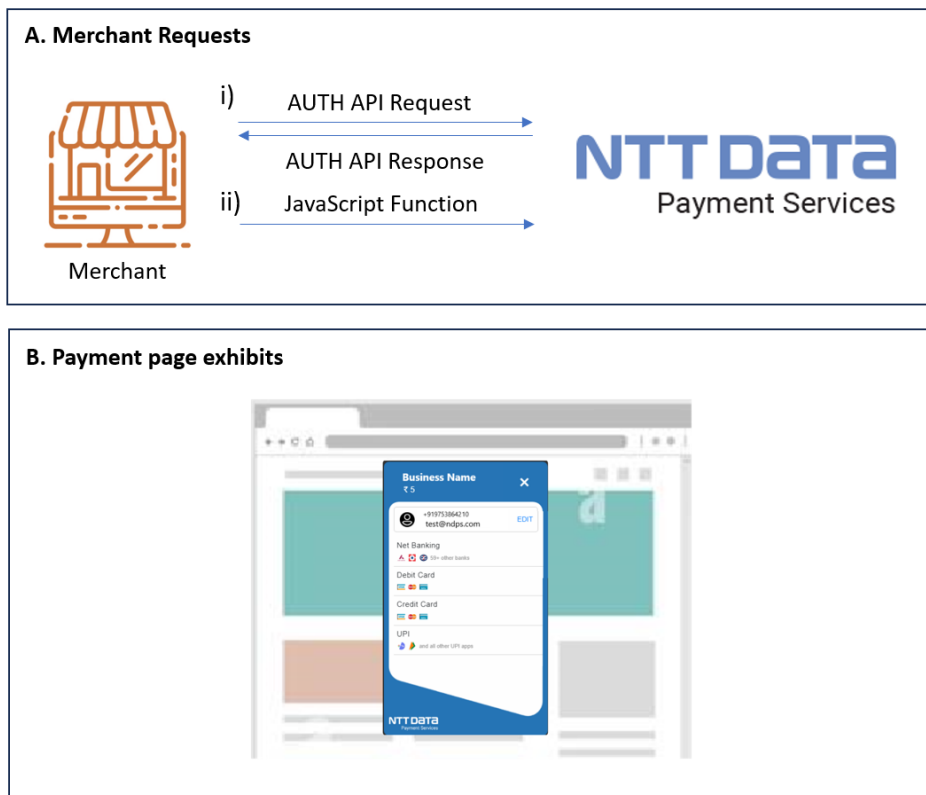
Below is the end-to-end payment process flow from –

- i) Once the customer is at merchant's checkout page, the merchant calls the NDPS' Rest API for payment authentication
- ii) Rest API request needs to be encrypted using the AES-512 algorithm
- iii) Token ID will be generated by NDPS and sent in response to the AUTH API request
- iv) The response will also be encrypted using the AES-512 algorithm
- v) The merchant is required to decrypt this response and thereby acquire the token ID
- vi) This token ID is to be used for calling the NDPS' JavaScript function which will then open the payment page
- vii) End customer can edit the mobile number and email ID which was sent by the merchant in the Rest API request and then continue to make the payment
- viii) Payment can be done by the customer using any of UPI, Net Banking, Credit card, Debit card, NEFT/RTGS or Wallet as the payment option
- ix) Once the payment is done, NDPS will send the response to the merchant entailing the status of the transaction (success, failed or timeout)

- x) In case if the token ID is not received by the merchant or the response is not received in the specified time-period then the error message will be shown to the merchant i.e., "Endpoint request timed out"}
- xi) Such errors are to be handled explicitly by the merchant and Rest API request has to be reinitiated.

## 4. Transaction APIs

**Process Flow :** Two API calls need to be triggered simultaneously for the exhibition of the payment page on merchant's website. The details of the same can be found below -



### a) AUTH API

- i) Merchant is required to initiate the **Rest API** to initiate payment link creation.
- ii) Authentication API used to generate the **Token ID**
- iii) Post this, the Token ID will be used to call the JavaScript function to then land on the Payment pop-up.
- iv) **Token ID** generation will be server-to-server using **POST** method.
- v) The request and response shall be encrypted and decrypted using **AES -512** encryption algorithm and a signature generated using **HMACSHA-512** algorithm.

## AUTH API Request

### Request Parameters :

| Parent JSON field Name | Parameter Name | Data Type | Max Length | Mandatory / Optional / Conditional | Sample Value                                                   | Static / Dynamic | Description                                                                  |
|------------------------|----------------|-----------|------------|------------------------------------|----------------------------------------------------------------|------------------|------------------------------------------------------------------------------|
| merchDetails           | merchId        | number    | 10         | M                                  | 330703                                                         | Static           | NDPS Merchant MID                                                            |
|                        | userId         | string    | 45         | O                                  | "123"                                                          | Dynamic          | Customer User ID                                                             |
|                        | password       | string    | 80         | M                                  | "cf7b79f6"                                                     | Static           | Transaction Password provide by NDPS                                         |
|                        | merchTxnId     | string    | 50         | M                                  | "Test123450"                                                   | Dynamic          | Unique Transaction Reference No. provided by Merchant                        |
|                        | merchTxnDate   | datetime  | 19         | M                                  | "2023-07-13 20:46:00"                                          | Dynamic          | Transaction date in format -> YYYY-MM-DD hh:mm:ss                            |
| payDetails             | amount         | number    | 12,2       | M                                  | 1.00                                                           | Dynamic          | Total Transaction Amount (10 digits prior to decimal)                        |
|                        | product        | string    | 50         | M                                  | "NCA"                                                          | Dynamic          | Unique value provided by NDPS pertains to merchant settlement account number |
|                        | custAccNo      | string    | 45         | C                                  | "213232323"                                                    | Dynamic          | End customer's account number (Only Mandatory for TPV Transaction)           |
|                        | txnCurrency    | string    | 5          | M                                  | "INR"                                                          | Static           | Transaction Currency Type                                                    |
| custDetails            | custEmail      | string    | 100        | M                                  | <a href="mailto:testuser@atomtech.in">testuser@atomtech.in</a> | Dynamic          | End customer's email ID                                                      |

|                     |            |            |      |   |              |         |                                                                                      |
|---------------------|------------|------------|------|---|--------------|---------|--------------------------------------------------------------------------------------|
|                     | custMobile | string     | 20   | M | "9753864210" | Dynamic | End customer's mobile number                                                         |
| extras              | udf1       | string     | 45   | O | -            | Dynamic | User Defined Parameter 1                                                             |
|                     | udf2       | string     | 45   | O | -            | Dynamic | User Define Parameter 2                                                              |
|                     | udf3       | string     | 45   | O | -            | Dynamic | User Define Parameter 3                                                              |
|                     | udf4       | string     | 45   | O | -            | Dynamic | User Define Parameter 4                                                              |
|                     | udf5       | string     | 45   | O | -            | Dynamic | User Define Parameter 5                                                              |
|                     | udf6       | string     | 255  | O | -            | Dynamic | User Define Parameter 6                                                              |
|                     | udf7       | string     | 255  | O | -            | Dynamic | User Define Parameter 7                                                              |
|                     | udf8       | string     | 255  | O | -            | Dynamic | User Define Parameter 8                                                              |
|                     | udf9       | string     | 255  | O | -            | Dynamic | User Define Parameter 9                                                              |
|                     | udf10      | string     | 255  | O | -            | Dynamic | User Define Parameter 10                                                             |
| payModeSpecificData | subChannel | string     | 40   | O | "NB"         | Dynamic | Use to exhibit specific payment mode                                                 |
| prodDetails         | prodName   | string [ ] | 50   | O | "NCA"        | Dynamic | Product name registered with NDPS (multiple product can be passed as an array) Refer |
|                     | prodAmount | number     | 12,2 | O | 1.00         | Dynamic | Product Specific Amount (10 digits prior to decimal)                                 |

The below are the Possible Values for Parameter 'subChannel' –

| Field Value | Description                                                                                |
|-------------|--------------------------------------------------------------------------------------------|
| NB          | In case the merchant wishes to show only <b>Net Banking</b> as payment option to customer. |
| CC          | In case the merchant wishes to show only <b>Credit Card</b> as payment option to customer. |
| DC          | In case the merchant wishes to show only <b>Debit Card</b> as payment option to customer.  |

|    |                                                                                                                |
|----|----------------------------------------------------------------------------------------------------------------|
| MW | In case the merchant wishes to show only <b>Wallet</b> as payment option to customer.                          |
| PP | In case the merchant wishes to show only <b>PhonePe</b> as payment option to customer.                         |
| PW | In case the merchant wishes to show only <b>Paytm Wallet</b> as payment option to customer.                    |
| EM | In case the merchant wishes to show only <b>EMI</b> as payment option to customer.                             |
| NR | In case the merchant wishes to show only <b>Challan</b> as payment option to customer.                         |
| BQ | In case the merchant wishes to show only <b>BharatQR</b> as payment option to customer.                        |
| UP | In case the merchant wishes to show only <b>Unified Payment Interface (UPI)</b> as payment option to customer. |

### Request in JSON format (Sample) :

```
{
  { "payInstrument": { "headDetails": { "version": "OTSv1.1", "api": "AUTH", "platform": "FLASH" },
  "merchDetails": { "merchId": "317157", "userId": "", "password": "Test@123", "merchTxnId": "019MC6VE42",
  "merchTxnDate": "2024-12-19 06:20:42" }, "payDetails": { "amount": "319", "product": "NSE", "custAccNo":
  "213232323", "txnCurrency": "INR", "clientCode": "" }, "custDetails": { "custFirstName": "Ravina Solse",
  "custEmail": "hrd-mum789@ai.in", "custMobile": "09754564348" }, "extras": { "udf1": "One", "udf2": "Two",
  "udf3": "Three", "udf4": "Four", "udf5": "Five", "udf6": "Six", "udf7": "Seven", "udf8": "Eight", "udf9": "Nine",
  "udf10": "Ten" }}}
  },
  "payModeSpecificData": {
    "subChannel": "DC"
  }
}
}
```

### Specifications for the Amount to be split to multiple Accounts :

Merchant can specify multiple products in 'prodDetails' if the same is pre-configured. The amounts across those products can be variable, but the total amount across all the products should be equal to that paid by the end-user.

### Process flow diagram :



If there are various products across which the payment is to be distributed, the 'product' parameter shall be replaced by the 'prodDetails' parent JSON field as depicted in the example below –



The "product": "NSE",' shall be replaced by :-

```
"prodDetails": [  
  {  
    "prodName": "NSE",  
    "prodAmount": 11.00  
  },  
  {  
    "prodName": "BSE",  
    "prodAmount": 12.00  
  }  
],
```

In such request, the sum of all 'prodAmount' should be equal to the amount to be paid.

### *Request in Encrypted format :*

- Merchant needs to encrypt the request using the AES -512 encryption algorithm.
- It is mandatory to pass the 'encData' and 'merchId' in the REST API call.

### *Sample Encrypted Request :*

FFC44EED0A1AE8C8FFDA9352895059AD02744AD2AA34BBDC0BDCD2680CF66E19873972E4F7571536A0D09A  
9FEC71A544E3489A49F2AB899A7F9C0A682419BF362DD2D89047F299B8E03DCC6BE49F963F852C25EB2ECEC  
54EFB4D59777831AF03A8BDC61D609C5CBF7D9BDF797451A4F07BB04323673A6E04E7F755243B113E06E3B4  
8B42230FE19A2C8D08EAB3D34BEF53D0D3EF9573DAA6D004280E71983B83E77824C5D0BDC0911C913DAD1  
79B4E75FB2260EA2CCAB12098ED5E2F53A357D11C6AB718076F73BFF167CB6D3D61B58214AE84EA0AF69D4  
A25F2D834767288957F20A3C6D92EE38B926E697A97BA2BE9EBBC53364B0E6864F17A4784055B2FFFFEDAA8  
F0D18C7ECBE5EA82D798558CA1C44B27882790A8184DA533F2149989D2D031A365711FF7D796F59DD2BD17  
E29B5D7734BD46D7A7106F47D381E79FAD0A5DD02BE3A5CEFA4BBBD4360294710D8CF7D9A458567E73DD2D  
CDD3F262A7DAF90CE8F574F1726651AB0C4DD6D3EC5C6EACFA393496A47CB0ACA0FD8299E817400E3A8433  
1AE307F468805B0257FC2D0192DDABAB7C7E1EDDA9D6DB1EE07F78A73BF2E2DACE2FD5CDB70A6B86F3FA7  
7D021D6A9030EAF9B5298B8418E211397DBC27ABF66AC004B7B13D8AC9C9F7057505283AD8C05D8DB10D3  
21370FACCB1C1B157B2CE7E0A8678793B3D57892E0F4662FA6FA2F8E72D3C6C17EA4CA49853BE79037ABD02  
2ADA162DAD4E5DE2F0349A9FB4D1F375DB637C6303CCEA7AEA9A43E25DA326E074B09C51AE6F3298727C6  
FBC18D8E92636F6BC9F1D90607988CE8B5E262B053890B62029B7E3E75970F9BC428D388CF990B6A65B588  
B1A7A8138C991471344785A414B70F97BE77D04ED813F5696DA97DB384208DE9F2151BFCD750A922F773B7  
50A612113A3267732DDD5

### **AUTH API Response**

#### *Response Parameters :*

| Parent JSON | Parameter Name | Data Type | Max Length | Sample Value | Static/ Dynamic | Description |
|-------------|----------------|-----------|------------|--------------|-----------------|-------------|
|-------------|----------------|-----------|------------|--------------|-----------------|-------------|

| field Name |                |        |     |                                                 |         |                                                 |
|------------|----------------|--------|-----|-------------------------------------------------|---------|-------------------------------------------------|
| encData    | atomTokenId    | number | 15  | 15000008670129                                  | Dynamic | Transaction Specific Token ID Generated by NDPS |
|            | txnMessage     | string | 20  | "SUCCESS"                                       | Dynamic | Status Message                                  |
|            | txnStatusCode  | string | 10  | "OTS0000"                                       | Dynamic | Status Code                                     |
|            | txnDescription | string | 100 | "ATOM TOKEN ID HAS BEEN GENERATED SUCCESSFULLY" | Dynamic | Status Description                              |

### *Sample Encrypted Response :*

merchId=330703&encData=67D1988C3CB9D7BAE404AD775CAD04C7CF51F4097AD369F73CD223B5A370328A53BED3977D2EB34EFE0692D8001950B37FD22D1A79992BBB7B7DA5D51509FEC08BA160760758718B7CB1679C61B8C3F9BCD2276DD306FDBBBFB26DAC7CF13D0E2C638E774D78BC418CF36B383371FE7ECB945AB8A505843DAF91861FD2D961F3DA42791E77C364A350BB5D48BAF864B53E24F07E2E986E7AF0AD1B33C73779C1E8A3EDC36C62ED3156B1B7F32C6C27D9

### *Response in JSON format :*

```
encData [Plain] :
{
  "atomTokenId":15000000033303, "responseDetails":{ "txnStatusCode":"OTS0000",
  "txnMessage":"SUCCESS",
  "txnDescription":"ATOM TOKEN ID HAS BEEN GENERATED SUCCESSFULLY"
}
}
```

Possible status code values and messages that will be posted in response while generating the Atom Token Id are mentioned below:

| S. No. | txnStatus | txnMessage | txnDescription                   |
|--------|-----------|------------|----------------------------------|
| 1      | OTS0000   | SUCCESS    | ATOM TOKEN ID HAS BEEN GENERATED |
| 2      | OTS0451   | FAILED     | INVALID MERCHANT INFORMATION     |
| 3      | OTS0600   | FAILED     | AUTH SERVICE FAILED              |
| 4      | OTS0600   | FAILED     | VALIDATION FAILED                |
| 5      | OTS0600   | FAILED     | TOKEN GENERATION FAILED          |

### *Response in Encrypted format :*

- Merchant needs to decrypt the response received using the AES-512 decryption algorithm.
- Post decryption of the 'atomTokenId' passed by NDPS in the response will be received by the merchant.
- For Production:  
merchId=330703&encData=67D1988C3CB9D7BAE404AD775CAD04C79FDBB8758C09EB33B62BD425EBD4FC1E4799D3F29BFE5A49AC8BFD759CFAA0EAB610F05B4EA962790E9E20BFD9BB0B116CB56290C1083E3ECAD9A25AFF201D06DB613402E6B42C73C6FB292D9B38822200A554AE9BE10760581F1CE388D96FBB08BDFD4F89BD95145AE3F9E64AC709058EECB56DC75F55464E74CFBC651EA28482E17EB7999C763A50967A96B205A6C34DB9C79A4EF58A1237DFE2BEBEF5D2F3

### b) JavaScript

- Once the 'atomTokenId' is received by the merchant, the JavaScript function needs to be called by the merchant.
- This JavaScript function can be called using the button click
- Mandatory JavaScript CDN link for UAT:  
<https://pgtest.atomtech.in/staticdata/ots/js/atomcheckout.js>
- Mandatory JavaScript CDN link for Production:  
<https://psa.atomtech.in/staticdata/ots/js/atomcheckout.js>

### JAVA Script in JSON format

```
<script > function openPay()
{
    const options =
    {
        "atomTokenId": "atomTokenId",
        "merchId": "8952",
        "custEmail": "testuser@ndps.in",
        "custMobile": "8888888888",
        "returnUrl": "http://your-response-url/response.php"
    }
    let atom = new AtomPaynetz(options, 'uat');
}
</script>
```

### c) Final Transaction Response

- Once the transaction is completed by the end user the final transaction response will be sent to the merchant by NDPS

- Transaction response will provide the details such as the payment mode used, Bank Name, Bank Transaction ID, Masked Card Number etc.
- Merchant can capture the details and accordingly show the response to the end customer.

### Response Parameters

All the below parameters are enclosed in the Parent JSON Header – ‘payInstrument’

| Parent JSON field Name | Parent JSON field Name | Parameter Name  | Data Type | Length | Description                                     | Mandatory / Optional |
|------------------------|------------------------|-----------------|-----------|--------|-------------------------------------------------|----------------------|
| merchDetails           |                        | merchId         | number    | 15     | Merchant Unique Reference Number shared by NDPS | M                    |
|                        |                        | merchTxnId      | string    | 50     | Transaction Id pertaining to merchant           | M                    |
|                        |                        | mccCode         | string    | 7      | Master Category Code provided by NDPS           | M                    |
|                        |                        | merchTxnDate    | datetime  | 19     | Transaction Date                                | M                    |
| payDetails             |                        | atomTxnID       | number    | 25     | NDPS Transaction ID                             | M                    |
|                        | prodDetails []         | prodName        | string    | 50     | Name of the Product                             | M                    |
|                        |                        | prodAmount      | number    | 12,2   | Product Specific Amount                         | M                    |
|                        |                        | Amount          | number    | 12,2   | <a href="#">Amount</a>                          | M                    |
|                        |                        | surchargeAmount | number    | 12,2   | Not to be passed                                | M                    |
|                        |                        | totalAmount     | number    | 12,2   | <a href="#">Total Amount</a>                    | M                    |
|                        |                        | custAccNo       | String    | 45     | Bank Account Number of end user                 | C (only for TPV)     |
|                        |                        | txnInitDate     | datetime  | 19     | NDPS Txn Initiated Date                         | M                    |
|                        |                        | txnCompleteDate | datetime  | 19     | Transaction Complete Date                       | M                    |
|                        |                        | custAcclfsc     | String    | 45     | IFSC of end user                                |                      |
|                        |                        | txnCurrency     | String    | 5      | Currency used for transaction                   | M                    |
|                        |                        | Signature       | String    | 256    | Signature generated                             | M                    |
|                        |                        | Remarks         | String    | 20     | Remark passed by Merchant                       | M                    |
|                        |                        | clientCode      | String    | 45     | Unique Client Code passed by Merchant           | M                    |
| custDetails            |                        | custFirstName   | String    | 50     | Name received in response from Issuing Bank     | M                    |
|                        |                        | custLastName    | String    | 50     | Last Name of User                               | M                    |
|                        |                        | custEmail       | Email     | 100    | Email of end user                               | M                    |

|                     |             |                |                                 |           |                              |                |
|---------------------|-------------|----------------|---------------------------------|-----------|------------------------------|----------------|
|                     | billingInfo | custMobile     | String                          | 45        | Mobile of End User           | M              |
|                     |             | custAddr1      | String                          | 200       | Customer Address Line 1      | M              |
|                     |             | custAddr2      | String                          | 200       | Customer Address Line 1      | M              |
|                     |             | custCountry    | String                          | 45        | Country                      | M              |
|                     |             | custCity       | String                          | 30        | City                         | M              |
|                     |             | custState      | String                          | 30        | State                        | M              |
|                     |             | custZipCode    | String                          | 10        | Zipcode                      | M              |
| payModeSpecificData | bankDetails | Subchannel []  | String                          | 40        | Sub Channel Id               | M              |
|                     |             | otsBankId      | String                          | 10        | Bank ID stored at Bank's end | M              |
|                     |             | otsBankName    | String                          | 60        | Sponsor Bank Name            | M              |
|                     |             | cardType       | String                          | 20        | CC/DC                        | CC-M DC-M      |
|                     |             | cardMaskNumber | String                          | 20        | Masked Card Number           | CC-M DC-M      |
|                     |             | qrString       | String                          | 200       | QR String                    | O              |
|                     |             | bankTxnId      | String                          | 30        | Bank Reference Id            | M              |
|                     |             | authId         | String                          | 10        | Bank authId                  | CC-M DC-M UP-M |
| scheme              | String      | 15             | Scheme Name (VISA, MASTER etc.) | CC-M DC-M |                              |                |
| responseDetails     |             | Message        | String                          | 30        | Success/Failure Message      | M              |
|                     |             | Description    | String                          | 50        | Description of Transaction   | M              |
|                     |             | statusCode     | String                          | 30        | Status Code of Transaction   | M              |
| extras              |             | udf1           | string                          | 45        | User Defined Parameter 1     | O              |
|                     |             | udf2           | string                          | 45        | User Define Parameter 2      | O              |
|                     |             | udf3           | string                          | 45        | User Define Parameter 3      | O              |
|                     |             | udf4           | string                          | 45        | User Define Parameter 4      | O              |
|                     |             | udf5           | string                          | 45        | User Define Parameter 5      | O              |

|  |  |       |        |     |                          |   |
|--|--|-------|--------|-----|--------------------------|---|
|  |  | udf6  | string | 255 | User Define Parameter 6  | O |
|  |  | udf7  | string | 255 | User Define Parameter 7  | O |
|  |  | udf8  | string | 255 | User Define Parameter 8  | O |
|  |  | udf9  | string | 255 | User Define Parameter 9  | O |
|  |  | udf10 | string | 255 | User Define Parameter 10 | O |

### Response in JSON format

```
{
  "payInstrument": {
    "merchDetails": {
      "merchId": "317157",
      "merchTxnId": "019MC6VE42",
      "merchTxnDate": "2024-12-19T11:56:37",
      "payDetails": {
        "atomTxnId": "11000000615892",
        "prodDetails": {
          "prodName": "NSE",
          "prodAmount": "319.00"
        },
        "amount": "319.00",
        "surchargeAmount": "17.67",
        "totalAmount": "336.67",
        "custAccNo": "213232323",
        "clientCode": "1234",
        "txnCurrency": "INR",
        "signature": "652d8df185e4073aa000f51ca2667f0e00458c4470061b1703e42981bb28887fb36145fb60b3cb3b3ab8f26d40c5a7b8019fac462101c436ee1c8e2e0731c339",
        "txnInitDate": "2024-12-19 11:56:41",
        "txnCompleteDate": "2024-12-19 11:57:21"
      },
      "payModeSpecificData": {
        "subChannel": "DC"
      },
      "bankDetails": {
        "otsBankId": "2",
        "bankTxnId": "00110000006158921119",
        "authId": "00000",
        "otsBankName": "Hdfc Bank",
        "cardType": "DC",
        "cardMaskNumber": "XXXXXXXXXXXX4444",
        "scheme": "VISA"
      },
      "extras": {
        "udf1": "One",
        "udf2": "Two",
        "udf3": "Three",
        "udf4": "Four",
        "udf5": "Five"
      },
      "custDetails": {
        "custFirstName": "Ravina Solse",
        "custEmail": "hrd-mum789@ai.in",
        "custMobile": "09754564348",
        "billingInfo": {}
      },
      "responseDetails": {
        "statusCode": "OTS0000",
        "message": "SUCCESS",
        "description": "SUCCESS"
      }
    }
  }
}
```

Possible status code values, messages and description that will be posted in the final transaction response are mentioned below :

| CODE    | MESSAGE | DESCRIPTION                                                   | DESCRIPTION |
|---------|---------|---------------------------------------------------------------|-------------|
| OTS0000 | SUCCESS | TRANSACTION SUCCESSFUL                                        |             |
| OTS0601 | FAILED  | IN STAGE TWO TRANSACTION, ATOM TXN ID SHOULD NOT BE NULL.     |             |
| OTS0602 | FAILED  | INCORRECT SURCHARGE AMOUNT.                                   |             |
| OTS0603 | FAILED  | SUBCHANNEL CAN NOT BE EMPTY.                                  |             |
| OTS0604 | FAILED  | IF SUBCHANNEL IS NB THEN BANK ID SHOULD NOT BE EMPTY.         |             |
| OTS0605 | FAILED  | IF SUBCHANNEL IS CC or DC THEN CARD DATA SHOULD NOT BE EMPTY. |             |
| OTS0606 | FAILED  | IF SUBCHANNEL IS UP THEN VPA SHOULD NOT BE EMPTY.             |             |

|         |        |                                                                                     |                                                |
|---------|--------|-------------------------------------------------------------------------------------|------------------------------------------------|
| OTS0607 | FAILED | IF SUBCHANNEL IS EM THEN EMI DETAILS SHOULD NOT BE EMPTY.                           |                                                |
| OTS0608 | FAILED | IN STAGE TWO TRANSACTION, SUBCHANNEL CAN NOT BE EMPTY.                              |                                                |
| OTS0609 | FAILED | CARD DETAILS ARE MISSING.                                                           |                                                |
| OTS0610 | FAILED | INVALID EMI TENURE.                                                                 |                                                |
| OTS0611 | FAILED | INVALID EMI BANK NAME.                                                              |                                                |
| OTS0612 | FAILED | SUMMATION OF AMOUNT & SURCHARGE AMOUNT SHOULD BE EQUAL TO TOTAL AMOUNT.             |                                                |
| OTS0613 | FAILED | RESPONSE IS ALREADY AVAILABLE IN DB FOR ATOM TXN ID.                                |                                                |
| OTS0614 | FAILED | ATOM's AMOUNT MISMATCHED WITH BANK's AMOUNT.                                        |                                                |
| OTS0615 | FAILED |                                                                                     | ANY OF THE MANDATORY PARAMETER MUST BE MISSING |
| OTS0616 | FAILED | INSUFFICIENT MERCHANT INFORMATION.                                                  |                                                |
| OTS0617 | FAILED | IN STAGE TWO TRANSACTION, DB STAGE SHOULD BE 1.                                     |                                                |
| OTS0618 | FAILED | IN STAGE TWO TRANSACTION, PROVIDED AMOUNT SHOULD BE EQAUL TO DB AMOUNT.             |                                                |
| OTS0619 | FAILED | SUBCHANNEL & CARD TYPE MISMATCHED.                                                  |                                                |
| OTS0620 | FAILED | CARD SCHEME MISMATCHED.                                                             |                                                |
| OTS0621 | FAILED | AMOUNT SHOULD BE WITHIN A RANGE.                                                    |                                                |
| OTS0622 | FAILED | ACCOUNT DETAILS MANDATORY FOR PENNYDROP.                                            |                                                |
| OTS0623 | FAILED | IN SEAMLESS TRANSACTION, WHEN MERCHANT IS ON SURCHARGE THEN SURCHARGE AMOUNT SHOULD |                                                |
| OTS0624 | FAILED | TRANSACTION RESPONSE IS ALREADY AVAILABLE FOR ATOM TXN ID.                          |                                                |
| OTS0625 | FAILED | BANK TXN AMOUNT IS MISMATCHED WITH ATOM TXN AMOUNT.                                 |                                                |
| OTS0626 | FAILED | AMOUNT & TOTAL AMOUNT OF ALL PASSED PRODUCTS SHOULD BE EQUAL.                       |                                                |
| OTS0627 | FAILED | INAPPROPRIATE WAY TO CONSUME OTS PAYMENT SERVICE.                                   |                                                |

|         |        |                                                                                                              |                                                             |
|---------|--------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| OTS0628 | FAILED | ATOM TXN ID SHOULD NOT BE EMPTY/NULL IN BANK RESPONSE.                                                       |                                                             |
| OTS0629 | FAILED | IN SEMI-SEAMLESS TRANSACTION, IF IT IS CARD TRANSACTION THEN SURCHARGE AMOUNT SHOULD NOT BE NULL/EMPTY/ZERO. |                                                             |
| OTS0630 | FAILED | CARD TYPE MISMATCHED                                                                                         |                                                             |
| OTS0631 | FAILED | FAILED-INVALID CARD NUMBER.                                                                                  |                                                             |
| OTS0632 | FAILED | FAILED-CARD EXP MONTH/YEAR SHOULD BE IN PROPER FORMAT.                                                       |                                                             |
| OTS0633 | FAILED | FAILED-CARD IS EXPIRED.                                                                                      |                                                             |
| OTS0634 | FAILED | FAILED-INVALID CVV.                                                                                          |                                                             |
| OTS0635 | FAILED | FAILED-TRANSACTION MODE NOT FOUND IN URL PATTERN.                                                            |                                                             |
| OTS0636 | FAILED | FAILED-UNABLE TO DECRYPT CARD DETAILS                                                                        |                                                             |
| OTS0637 | FAILED | FAILED-UNABLE TO DECRYPT ENC DATA                                                                            |                                                             |
| OTS0638 | FAILED |                                                                                                              | ANY OF THE MANDATORY PARAMETER MUST BE MISSING FROM ENCDATA |
| OTS0639 | FAILED | FAILED-FUTURE DATE NOT ALLOWED.                                                                              |                                                             |
| OTS0640 | FAILED | FAILED-DATE FORMAT MUST BE "+OTSUtil.ATOM_DATE_PATTERN";                                                     |                                                             |
| OTS0641 | FAILED | MERCHID OR ENCDATA MISSING.                                                                                  |                                                             |

## Signature Generation

- For above mentioned API's request and their response signature fields to be generated using the shared hashing code.
- Signature type is HMACSHA512.

String to be encrypted:-

[ merchId + atomTxnId + merchTxnId + totalAmount + txnStatusCode + subChannel + bankTxnId ]

### a) Sample Request

1191110000001234abc12310.00OTS0000NB11dsff232311

### b) Generated Signature



fb0798ae1602bc0d395b7a12e8f08ab5d0ec64e90a2b9e7f09ed3db847ac86820edb1a0d557c51aec2e0166fae9  
7d4aa966a6f0d8512fd454e4deb906959c1bf

### c) Hash Keys

- Below are the merchant Id, request & response hash key –

| MerchId | reqHashKey    | respHashKey       |
|---------|---------------|-------------------|
| 9135    | KEY1234567234 | KERESPY1234567234 |

### d) Encryption Key

- Below are the UAT merchant id, request & response encryption key –

| MerchId | encReqKey                        | encResKey                        |
|---------|----------------------------------|----------------------------------|
| 9135    | 58BE879B7DD635698764745511C704AB | 7813E3E5E93548B096675AC27FE2C850 |

### e) Signature Generation Java Code:

```
import java.io.PrintStream;
import java.io.UnsupportedEncodingException;
import java.security.InvalidKeyException;
import java.security.Key;
import java.security.NoSuchAlgorithmException;
import javax.crypto.Mac;
import javax.crypto.spec.SecretKeySpec;

public class AtomSignature {
    public static String generateSignature(String hashKey, String[] param) {
        String resp = null;
        StringBuilder sb = new StringBuilder();
        for (String s: param) {
            sb.append(s);
        }

        try {
            System.out.println("String =" + sb.toString());
            resp = byteToHexString(encodeWithHMACSHA2(sb.toString(), hashKey));
        } catch (Exception e) {
            System.out.println("Unable to encod value with key :"+ hashKey + " and input :"+ sb.toString());
            e.printStackTrace();
        }
        return resp;
    }

    private static byte[] encodeWithHMACSHA2(String text, String keyString)
    throws NoSuchAlgorithmException, InvalidKeyException, UnsupportedEncodingException {
        Key sk = new SecretKeySpec(keyString.getBytes("UTF-8"), "HMACSHA512");
        Mac mac = Mac.getInstance(sk.getAlgorithm());
        mac.init(sk);
```

```

        byte[] hmac = mac.doFinal(text.getBytes("UTF-8"));
        return hmac;
    }
    public static String byteToHexString(byte byData[]) {
        StringBuilder sb = new StringBuilder(byData.length * 2);
        for (int i = 0; i < byData.length; i++) {
            int v = byData[i] & 0xff;
            if (v < 16)
                sb.append('0');
            sb.append(Integer.toHexString(v));
        }
        return sb.toString();
    }
}

```

## Response in Encrypted format

- Final Transaction response will be sent by NDPS in the encrypted format
- Merchant needs to decrypt the response using AES-512 decryption algorithm

## Sample Encrypted Response :

merchId=8952&encData=0CD63904F407DC4D24167F8D8F2BEE997C80D4536815B71C1EDDB8FDCBACE8B9F6D72174CC18263D88AD7728C821A1F7B789A3F6C8EE43C28933A6748D3239346F4AAEC04AFFD7687FB31FFDA9AA F0F64A2911990F315B425F11F1196EAF59E9471644C321A0050BF64C72DA0EFE6A11D14F4AA3890267552FA4E3F A84B21275150F7E98CCF84BD145DCB6F00EF7494FAEE9D23852507F46517B67D8F9680941BF887A761D9B26879 B2A369E9B6CEFF835DD51DEE86982255CCB98A8C9D8C3662FB87AF859B9E0DF167D7D48E89BE6896226C5D8A ED6B2CA1986325D030B646ADAD4C3944A8B851D3D3448C7F0CD37DB8E466A51AFFD43E38307E2ECA2967561 4AC4DAA072CB6595974E4E1EEEE70F031B7BA3C5FDF2D4F599854B658B9D9D2DAA3FEAA67B534BFAD98F B5D123BB8FDB563F840B7B28C71054CCA358F50348CE842C3AA05FDBF0400892F03B5290FF5CA7E01C174FCE BC0B65012FF9F0AB1C19F483570599F9C681001F46C342CD6A491B2B4619FC70D30FF0CEAB49F2B5201ADCE25 C510027496A4A64EA3A2661D55DA89741600 9769F2F501CE864F8360EC8EB7558CB24AAC577E338EA3DB515DA5199D159688C0A624D0C548135A6001D2543 D7274BEBA6C2AB648042C08CDF4732CB63FB0B57DD967E8F577427EF19D669E0A16AF77CA605491C1657744 A1395065C1A459232A12743E62DEF8B6FC9F6075C8DA7E27C2AE6874698FAD367A9C23CB28FDC001AE5501693 FB5435BD871DDCC5BB0E236867F60AA04E52A7582DB8F25D0FC4CFD7C4D223D67DADF248E0C920F0BCA7C9 E774604FAEB56FAE025091FA595A7797C4331D3FCF9CEB881A1392F231A4F05F4A4FFA58099802A20F7345EC71 20BCE82ACD7C536B2627E4DD22F2CE177C27F38B544ED991EF83A1F888A6217B48511E6A1FCFEF065FD84C459 E4695E39AD5BB078B42751A798B780258E64B2DA311A782D3B6FCD987515E870A54465F7C5302660B6F3A0CF88 908EDE72743FBD55741B8D7B9C6BBAE1A9D6F174DC4A9C52EB8129A8DE69F49A340BA65AF128E935FED44D5 7906B274905E6297C00358FBEE401742C5812C9BE54517770F48143B9E8E9EA39617834968AAC4A1494E3D5CF3B 69BD5D1ED0392CF84B86C64E4B29CDCB81C7FEA1ADEA09B9C27D6F31A5DB9EE6A27F1E0914D2180D5A6247 A5BE6F2119FA13919DC696B0C4EA95E5D96F850CB6D32B05F5505F48E2354F4953A2D2897F929D346ECC39F816 DBF4079893333A44AFA473B4A86FF96773295B48BA192FAAFEF79126FE0E944E1B1CE0D09C2BC745F62C237C3 E57EC61209ACF426E8FAB1DB3F8DDDE7AC6374866964178

## Sample Decrypted Response :

```

{
  "payInstrument": {
    "merchDetails": {

```

```
"merchId": 8952,
"merchTxnId": "Test123450",
"merchTxnDate": "2021-12-06 T11:40:18"
},
"payDetails": {
  "atomTxnId": 11000000174925,
  "prodDetails": [
    {
      "prodName": "NSD",
      "prodAmount": 100.00
    }
  ],
  "amount": 100.00,
  "surchargeAmount": 1.18,
  "totalAmount": 101.18,
  "custAccNo": "213232323",
  "clientCode": "1234",
  "txnCurrency": "INR",
  "signature":
    "2b12c8bfc0e3a8268eddb6f406bf4187d4d0a0064d0355446986511453922c27e38367a97fff85863d48c147a8218e9e2d5003ab121f6f61ce3914030c60caac",
  "txnInitDate": "2021-12-06 11:40:19",
  "txnCompleteDate": "2021-12-06 11:40:23"
},
"payModeSpecificData": {
  "subChannel": [
    "NB"
  ],
  "bankDetails": {
    "otsBankId": 2001,
    "bankTxnId": "vYmy8VmFzxLF6A7uDx0H",
    "otsBankName": "Atom Bank"
  }
},
"extras": {
  "udf1": "",
  "udf2": "",
  "udf3": "",
  "udf4": "",
  "udf5": "",
  "udf6": "",
  "udf7": "",
  "udf8": "",
  "udf9": "",
  "udf10": ""
},
"custDetails": {
  "custEmail": "testuser@nttdata.com",
  "custMobile": "9753864210",
```

```

    "billingInfo": {}
  },
  "responseDetails": {
    "statusCode": "OTS0000",
    "message": "SUCCESS",
    "description": "TRANSACTION IS SUCCESSFUL."
  }
}
}
}

```

**Note\*** : In case of NEFT/RTGS challan transaction, merchant will receive real-time transaction status as "Pending" as 1st stage when the Challan is generated successfully.  
 Post the payment as per NEFT/RTGS challan and reconciliation from NDPS, response will be posted only on Call Back URL (Webhook URL) configured against the Merchant MID. Hence merchants must have a Call Back URL compulsorily configured against their respective MID to receive the final transaction response.

## II. AES Encryption Logic

- Transaction Status (Requery) API's request and returned response are shared via AES-512 encryption.
- The following KEY are to be used for UAT:

| MerchId | encResKey                        | encReqKey                        |
|---------|----------------------------------|----------------------------------|
| 9135    | 58BE879B7DD635698764745511C704AB | 7813E3E5E93548B096675AC27FE2C850 |

### AES Encryption Java Code:

```

import java.util.logging.Logger;
import javax.crypto.Cipher;
import javax.crypto.SecretKey;
import javax.crypto.SecretKeyFactory;
import javax.crypto.spec.IvParameterSpec;
import javax.crypto.spec.PBEKeySpec;
import javax.crypto.spec.SecretKeySpec;

public class AtomEncryption {
    static Logger log = Logg\
er.getLogger(AtomEncryption.class.getName());
    private static int pswdIterations = 65536;
    private static int keySize = 512;
    private static final byte[] ivBytes = {
        0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15
    };

    public static String encrypt(String plainText, String key) {
        try {
            byte[] saltBytes = key.getBytes("UTF-8");

            SecretKeyFactory factory = SecretKeyFactory.getInstance("PBKDF2WithHmacSHA512");
            PBEKeySpec spec = new PBEKeySpec(key.toCharArray(), saltBytes, pswdIterations, keySize);

            SecretKey secretKey = factory.generateSecret(spec);
            SecretKeySpec secret = new SecretKeySpec(secretKey.getEncoded(), "AES");

```

```

        IvParameterSpec locallyIvParameterSpec = new IvParameterSpec(ivBytes);
        Cipher cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");
        cipher.init(1, secret, locallyIvParameterSpec);

        byte[] encryptedTextBytes = cipher.doFinal(plainText.getBytes("UTF-8"));

        return byteToHex(encryptedTextBytes);
    } catch (Exception e) {
        log.info("Exception while encrypting data:" + e.toString());
    }
    return null;
}

public static String decrypt(String encryptedText, String key) {
    try {
        byte[] saltBytes = key.getBytes("UTF-8");
        byte[] encryptedTextBytes = hex2ByteArray(encryptedText);

        SecretKeyFactory factory = SecretKeyFactory.getInstance("PBKDF2WithHmacSHA512");
        PBEKeySpec spec = new PBEKeySpec(key.toCharArray(), saltBytes, pswdIterations, keySize);
        SecretKey secretKey = factory.generateSecret(spec);
        SecretKeySpec secret = new SecretKeySpec(secretKey.getEncoded(), "AES");
        IvParameterSpec locallyIvParameterSpec = new IvParameterSpec(ivBytes);
        Cipher cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");
        cipher.init(2, secret, locallyIvParameterSpec);
        byte[] decryptedTextBytes = (byte[]) null;
        decryptedTextBytes = cipher.doFinal(encryptedTextBytes);
        return new String(decryptedTextBytes);
    } catch (Exception e) {
        log.info("Exception while decrypting data:" + e.toString());
    }
    return null;
}

private static String byteToHex(byte[] byData) {
    StringBuffer sb = new StringBuffer(byData.length * 2);
    for (int i = 0; i < byData.length; ++i) {
        int v = byData[i] & 0xFF;
        if (v < 16)
            sb.append('0');
        sb.append(Integer.toHexString(v));
    }
    return sb.toString().toUpperCase();
}

private static byte[] hex2ByteArray(String sHexData) {
    byte[] rawData = new byte[sHexData.length() / 2];
    for (int i = 0; i < rawData.length; ++i) {
        int index = i * 2;
        int v = Integer.parseInt(sHexData.substring(index, index + 2), 16);
        rawData[i] = (byte) v;
    }
    return rawData;
}

public static void main(String[] args) {
    try {
        String encryptedData = AtomEncryption.encrypt("1235", "ASWKLSLLFS4sd4g4gsdg");
        System.out.println("encryptedData : " + encryptedData);
    } catch (Exception e) {

```

```

        }
    }
}

```

```

// TODO: handle exception

```

## UAT environment details:

The UAT environment details are as follows :

**UAT IP :** [13.127.25.237](https://13.127.25.237)

**UAT server :** The UAT server needs to be whitelisted at the merchant's end so that response can be posted.

## Test Credentials

| UAT Test Credentials |                                               | Remarks                                                   |
|----------------------|-----------------------------------------------|-----------------------------------------------------------|
| Payment Domain URL   | https://paynetzuat.atomtech.in/ots/aipay/auth |                                                           |
| merchId              | 317159                                        | Alternatively, can use MIDs – 317156, 317157, 8952        |
| password             | Test@123                                      | Password                                                  |
| product/prodName     | NSE / AIPAY                                   | Product Name                                              |
| Request Key/Salt     | A4476C2062FFA58980DC8F79EB6A799E              | Encryption Key                                            |
| Response Key/Salt    | 75AEF0FA1B94B3C10D4F5B268F757F11              | Decryption Key                                            |
| Request Hash Key     | KEY123657234                                  | Used for signature validation in request (if applicable)  |
| Response Hash Key    | KEYRESP123657234                              | Used for signature validation in response (if applicable) |

## Test Card Details

| Test - Credit Card |                  |
|--------------------|------------------|
| Card Number:       | 4012888888881880 |
| Expiry:            | 12/23            |
| Cvv                | 123              |

|            |      |
|------------|------|
| Card Name: | Test |
|------------|------|

| Test - Debit Card |                  |
|-------------------|------------------|
| Card Number:      | 5555555555554440 |
| Expiry:           | 12/24            |
| Cvv               | 123              |
| Card Name:        | Test             |

### Test UPI ID

| Test - UPI ID |                    | Important Note                             |
|---------------|--------------------|--------------------------------------------|
| UPI ID        | <u>atomots@upi</u> | Reach out prior to conducting transaction  |
| UPI Name      | Atom OTS Vpa       | Auto populated once the UPI ID is verified |

## Sample Encrypted Request & Response (Single Account Settlement)

### Plain Text Json:

```
{
  "payInstrument": {
    "headDetails": {
      "version": "OTSv1.1",
      "api": "AUTH",
      "platform": "FLASH"
    },
    "merchDetails": {
      "merchId": "317159",
      "userId": "",
      "password": "Test@123",
      "merchTxnDate": "2022-03-07 20:46:00",
      "merchTxnId": "test000123"
    },
    "payDetails": {
      "amount": "100",
      "product": "NSE",
      "custAccNo": "213232323",
      "txnCurrency": "INR"
    },
    "custDetails": {
      "custEmail": "user@atomtech.in",
      "custMobile": "9898989898"
    }
  },
}
```

```

"extras": {
  "udf1": "Merchant Data",
  "udf2": "Merchant Data",
  "udf3": "Merchant Data",
  "udf4": "Merchant Data",
  "udf5": "Merchant Data"
  "udf6": " Merchant Data "
  "udf7": " Merchant Data "
  "udf8": " Merchant Data "
  "udf9": " Merchant Data "
  "udf10": " Merchant Data "
}
}
}

```

### Netbanking Response Sample format

(Success):

```

{
  "payInstrument": {
    "merchDetails": {
      "merchId": 317159,
      "merchTxnId": "test000123",
      "merchTxnDate": "2023-05-30T15:08:40"
    },
    "payDetails": {
      "atomTxnId": 11000000379742,
      "prodDetails": [
        {
          "prodName": "NSE",
          "prodAmount": 100
        }
      ],
      "amount": 100,
      "surchargeAmount": 0,
      "totalAmount": 100,
      "custAccNo": "213232323",
      "clientCode": "1234",
      "txnCurrency": "INR",
      "signature":
"1cbf87634568eabe245df5edf793f80b3de7796cc83246360cbb355a4b80cef2bc8f6a92245e213f6f03190298
72d2056d4422f650e6efc2b306aaa46f8d46a6",
      "txnInitDate": "2023-05-30 15:08:40",
      "txnCompleteDate": "2023-05-30 15:08:45"
    },
    "payModeSpecificData": {
      "subChannel": [
        "NB"
      ],
      "bankDetails": {

```



```

    "otsBankId": 2001,
    "bankTxnId": "zNX8R7LZRZFMXba8YrRz",
    "otsBankName": "Atom Bank"
  }
},
"extras": {
  "udf1": "Merchant Data",
  "udf2": "Merchant Data",
  "udf3": "Merchant Data",
  "udf4": "Merchant Data",
  "udf5": "Merchant Data"
  "udf6": " Merchant Data "
  "udf7": " Merchant Data "
  "udf8": " Merchant Data "
  "udf9": " Merchant Data "
  "udf10": " Merchant Data "
},
"custDetails": {
  "custEmail": "user@atomtech.in",
  "custMobile": "9898989898",
  "billingInfo": {}
},
"responseDetails": {
  "statusCode": "OTS0000",
  "message": "SUCCESS",
  "description": "TRANSACTION IS SUCCESSFUL."
}
}
}

```

#### Credit Card Response Sample format:

(Success):

```

{
  "payInstrument": {
    "merchDetails": {
      "merchId": 317159,
      "merchTxnId": "test000123",
      "merchTxnDate": "2023-05-30T15:07:50"
    },
    "payDetails": {
      "atomTxnId": 11000000379740,
      "prodDetails": [
        {
          "prodName": "NSE",
          "prodAmount": 100
        }
      ],
      "amount": 100,
      "surchargeAmount": 0,

```

```

    "totalAmount": 100,
    "custAccNo": "213232323",
    "clientCode": "1234",
    "txnCurrency": "INR",
    "signature":
"aa05829a4f42689e4488d75a84f0f5658cd4edc5d2a33772e1d971f28f5d4b8fb5848a07de8c0a1754b92b7d0
bcc44e3765afc664b197131394fff5271e22317",
    "txnInitDate": "2023-05-30 15:07:53",
    "txnCompleteDate": "2023-05-30 15:08:01"
  },
  "payModeSpecificData": {
    "subChannel": [
      "CC"
    ],
    "bankDetails": {
      "otsBankId": 2,
      "bankTxnId": "0011000000379740430",
      "authId": "00000",
      "otsBankName": "Hdfc Bank",
      "cardType": "CC",
      "cardMaskNumber": "XXXXXXXXXXXX1881",
      "scheme": "VISA"
    }
  },
  "extras": {
    "udf1": "Merchant Data",
    "udf2": "Merchant Data",
    "udf3": "Merchant Data",
    "udf4": "Merchant Data",
    "udf5": "Merchant Data"
    "udf6": " Merchant Data "
    "udf7": " Merchant Data "
    "udf8": " Merchant Data "
    "udf9": " Merchant Data "
    "udf10": " Merchant Data "
  },
  "custDetails": {
    "custEmail": "user@atomtech.in",
    "custMobile": "9898989898",
    "billingInfo": {}
  },
  "responseDetails": {
    "statusCode": "OTS0000",
    "message": "SUCCESS",
    "description": "SUCCESS"
  }
}

```

#### Debit Card Response Sample format:

{Success}:

```

{
  "payInstrument": {
    "merchDetails": {
      "merchId": 317159,
      "merchTxnId": "test000123",
      "merchTxnDate": "2023-05-30T15:11:21"
    },
    "payDetails": {
      "atomTxnId": 11000000379749,
      "prodDetails": [
        {
          "prodName": "NSE",
          "prodAmount": 100
        }
      ],
      "amount": 100,
      "surchargeAmount": 0,
      "totalAmount": 100,
      "custAccNo": "213232323",
      "clientCode": "1234",
      "txnCurrency": "INR",
      "signature":
        "c08f7e125a0979507aadb7c37a1ff829de694a1515361707960b4810fe3366cbd120e151af3ebacf3fb1c769b58f
        0559bee96f333f6dd62faacf36eb4b3ca082",
      "txnInitDate": "2023-05-30 15:11:23",
      "txnCompleteDate": "2023-05-30 15:11:31"
    },
    "payModeSpecificData": {
      "subChannel": [
        "DC"
      ],
      "bankDetails": {
        "otsBankId": 2,
        "bankTxnId": "0011000000379749430",
        "authId": "00000",
        "otsBankName": "Hdfc Bank",
        "cardType": "DC",
        "cardMaskNumber": "XXXXXXXXXXXX4444",
        "scheme": "VISA"
      }
    },
    "extras": {
      "udf1": "Merchant Data",
      "udf2": "Merchant Data",
      "udf3": "Merchant Data",
      "udf4": "Merchant Data",
      "udf5": "Merchant Data"
      "udf6": " Merchant Data "
      "udf7": " Merchant Data "
      "udf8": " Merchant Data "
      "udf9": " Merchant Data "
      "udf10": " Merchant Data "
    },
  },

```

```

"custDetails": {
  "custEmail": "user@atomtech.in",
  "custMobile": "9898989898",
  "billingInfo": {}
},
"responseDetails": {
  "statusCode": "OTS0000",
  "message": "SUCCESS",
  "description": "SUCCESS"
}
}
}

```

### UPI Response Sample format:

(Success):

```

{
  "payInstrument": {
    "merchDetails": {
      "merchId": 317159,
      "merchTxnId": "test000123",
      "merchTxnDate": "2023-05-30T15:12:33"
    },
    "payDetails": {
      "atomTxnId": 11000000379751,
      "prodDetails": [
        {
          "prodName": "NSE",
          "prodAmount": 100
        }
      ],
      "amount": 100,
      "surchargeAmount": 0,
      "totalAmount": 100,
      "custAccNo": "213232323",
      "clientCode": "1234",
      "txnCurrency": "INR",
      "signature":
        "f61affbc4c253aff7a3f85d8fd5d7ca44c46736ef0ea4595f3d924893ff07e7c70c6561b18347b5698aa62d9f300
        8129fe391be9f80e6f73d073593b1bfd68c0",
      "txnInitDate": "2023-05-30 15:12:35",
      "txnCompleteDate": "2023-05-30 15:13:24"
    },
    "payModeSpecificData": {
      "subChannel": [
        "UP"
      ],
      "bankDetails": {
        "otsBankId": 2,
        "bankTxnId": "1685439751273",

```

```

    "authId": "2391685439751273",
    "otsBankName": "Hdfc Bank"
  }
},
"extras": {
  "udf1": "Merchant Data",
  "udf2": "Merchant Data",
  "udf3": "Merchant Data",
  "udf4": "Merchant Data",
  "udf5": "Merchant Data",
  "udf6": " Merchant Data "
  "udf7": " Merchant Data "
  "udf8": " Merchant Data "
  "udf9": " Merchant Data "
  "udf10": " Merchant Data "
},
"custDetails": {
  "custEmail": "user@atomtech.in",
  "custMobile": "9898989898",
  "billingInfo": {}
},
"responseDetails": {
  "statusCode": "OTS0000",
  "message": "SUCCESS",
  "description": "APPROVED OR COMPLETED SUCCESSFULLY"
}
}
}

```

#### NEFT/RTGS Challan Response Sample format:

```

{
  "payInstrument": {
    "merchDetails": {
      "merchId": 317159,
      "merchTxnId": "test000123",
      "merchTxnDate": "2023-05-30T15:14:02"
    },
    "payDetails": {
      "atomTxnId": 11000000379754,
      "prodDetails": [
        {
          "prodName": "NSE",
          "prodAmount": 100
        }
      ],
      "amount": 100,
      "surchargeAmount": 0,
      "totalAmount": 100,
      "custAccNo": "213232323",
      "clientCode": "1234",

```

```

    "txnCurrency": "INR",
    "signature":
"5252acc5b04a57a09e027d4986c8e2dc6dea46202e9c424e4555ad12ba40d3feb8333b34c96a9b1569670ed7
1e41596c07a44a69e775d46eb42e53800c112ea9",
    "txnInitDate": "2023-05-30 15:14:03",
    "txnCompleteDate": ""
  },
  "payModeSpecificData": {
    "subChannel": [
      "NR"
    ],
    "bankDetails": {
      "otsBankId": 16,
      "bankTxnId": "ZATOMZ000000379754",
      "authId": "ZATOMZ000000379754"
    }
  },
  "extras": {
    "udf1": "Merchant Data",
    "udf2": "Merchant Data",
    "udf3": "Merchant Data",
    "udf4": "Merchant Data",
    "udf5": "Merchant Data"
    "udf6": " Merchant Data "
    "udf7": " Merchant Data "
    "udf8": " Merchant Data "
    "udf9": " Merchant Data "
    "udf10": " Merchant Data "
  },
  "custDetails": {
    "custEmail": "user@atomtech.in",
    "custMobile": "9898989898",
    "billingInfo": {}
  },
  "responseDetails": {
    "statusCode": "OTS0551",
    "message": "PENDING",
    "description": "CHALLAN GENERATED SUCCESSFULLY"
  }
}

```

## Sample Encrypted Request & Response (Split Settlement – Multi Product)

### Plain Text Json:

```

{
  "payInstrument": {
    "headDetails": {
      "version": "OTSv1.1",

```

```

    "api": "AUTH",
    "platform": "FLASH"
  },
  "merchDetails": {
    "merchId": "317159",
    "userId": "",
    "password": "Test@123",
    "merchTxnDate": "2022-03-07 20:46:00",
    "merchTxnId": "test000123"
  },
  "payDetails": {
    "amount": "100.00",
    "product": "AIPAY",
    "custAccNo": "213232323",
    "txnCurrency": "INR"
  },
  "prodDetails": [
    {
      "prodName": "AIPAY",
      "prodAmount": "20.00"
    },
    {
      "prodName": "NSE",
      "prodAmount": "80.00"
    }
  ]
},
"custDetails": {
  "custEmail": "user@atomtech.in",
  "custMobile": "9898989898"
},
"extras": {
  "udf1": "Merchant Data",
  "udf2": "Merchant Data",
  "udf3": "Merchant Data",
  "udf4": "Merchant Data",
  "udf5": "Merchant Data",
  "udf6": " Merchant Data "
  "udf7": " Merchant Data "
  "udf8": " Merchant Data "
  "udf9": " Merchant Data "
  "udf10": " Merchant Data "
}
}

```

### Netbanking Response Sample format

(Success):

```

{
  "payInstrument": {

```

```

"merchDetails": {
  "merchId": 317159,
  "merchTxnId": "test000123",
  "merchTxnDate": "2023-05-30T15:45:44"
},
"payDetails": {
  "atomTxnId": 11000000379816,
  "prodDetails": [
    {
      "prodName": "AIPAY",
      "prodAmount": 100
    }
  ],
  "amount": 100,
  "surchargeAmount": 0,
  "totalAmount": 100,
  "custAccNo": "213232323",
  "clientCode": "1234",
  "txnCurrency": "INR",
  "signature":
"Obfac7fd1fc27660beec679f17da434305b7b05bb0e63641b850c078d63cd82f45b258c65e4bac165be64b743
20f0583574efc488299b275592110bfb3dda785",
  "txnInitDate": "2023-05-30 15:45:44",
  "txnCompleteDate": "2023-05-30 15:45:50"
},
"payModeSpecificData": {
  "subChannel": [
    "NB"
  ],
  "bankDetails": {
    "otsBankId": 2001,
    "bankTxnId": "NgvleGMKpD7BcNPSZMYn",
    "otsBankName": "Atom Bank"
  }
},
"extras": {},
"custDetails": {
  "billingInfo": {}
},
"responseDetails": {
  "statusCode": "OTS0000",
  "message": "SUCCESS",
  "description": "TRANSACTION IS SUCCESSFUL."
}
}
}

```

#### Credit Card Response Sample format:

(Success):



```
{
  "payInstrument": {
    "merchDetails": {
      "merchId": 317159,
      "merchTxnId": "test000123",
      "merchTxnDate": "2023-05-30T15:54:22"
    },
    "payDetails": {
      "atomTxnId": 11000000379836,
      "prodDetails": [
        {
          "prodName": "AIPAY",
          "prodAmount": 100
        }
      ],
      "amount": 100,
      "surchargeAmount": 0,
      "totalAmount": 100,
      "custAccNo": "213232323",
      "clientCode": "1234",
      "txnCurrency": "INR",
      "signature":
        "c2f352c8830cc039f51d1a0d916ef30fac73abce78c740bc8b37cd0f4c1459cbc35b3a9dc953c07c1496e40ae00
        6a3f2f9d4dc53049788f13189fbbd404187f8",
      "txnInitDate": "2023-05-30 15:54:24",
      "txnCompleteDate": "2023-05-30 15:54:32"
    },
    "payModeSpecificData": {
      "subChannel": [
        "CC"
      ],
      "bankDetails": {
        "otsBankId": 2,
        "bankTxnId": "0011000000379836430",
        "authId": "00000",
        "otsBankName": "Hdfc Bank",
        "cardType": "CC",
        "cardMaskNumber": "XXXXXXXXXXXX1881",
        "scheme": "VISA"
      }
    },
    "extras": {},
    "custDetails": {
      "billingInfo": {}
    },
    "responseDetails": {
      "statusCode": "OTS0000",
      "message": "SUCCESS",
      "description": "SUCCESS"
    }
  }
}
```

```
}  
}
```

#### Debit Card Response Sample format:

(Success):

```
{  
  "payInstrument": {  
    "merchDetails": {  
      "merchId": 317159,  
      "merchTxnId": "test000123",  
      "merchTxnDate": "2023-05-30T15:56:54"  
    },  
    "payDetails": {  
      "atomTxnId": 11000000379842,  
      "prodDetails": [  
        {  
          "prodName": "AIPAY",  
          "prodAmount": 100  
        }  
      ],  
      "amount": 100,  
      "surchargeAmount": 0,  
      "totalAmount": 100,  
      "custAccNo": "213232323",  
      "clientCode": "1234",  
      "txnCurrency": "INR",  
      "signature":  
"279e30ef2d212ba9a6c83544bc3533ef8b6747dad737a2070eb2643b8511c1a7a1000d09b11c6e9a2396794ff  
114702d2aee0757aeefd689f20e5a23f8753d25",  
      "txnInitDate": "2023-05-30 15:56:57",  
      "txnCompleteDate": "2023-05-30 15:57:04"  
    },  
    "payModeSpecificData": {  
      "subChannel": [  
        "DC"  
      ],  
      "bankDetails": {  
        "otsBankId": 2,  
        "bankTxnId": "0011000000379842430",  
        "authId": "00000",  
        "otsBankName": "Hdfc Bank",  
        "cardType": "DC",  
        "cardMaskNumber": "XXXXXXXXXXXX4444",  
        "scheme": "VISA"  
      }  
    },  
    "extras": {},  
    "custDetails": {
```

```

    "billingInfo": {}
  },
  "responseDetails": {
    "statusCode": "OTS0000",
    "message": "SUCCESS",
    "description": "SUCCESS"
  }
}

```

### UPI Response Sample format:

(Success):

```

{
  "payInstrument": {
    "merchDetails": {
      "merchId": 317159,
      "merchTxnId": "test000123",
      "merchTxnDate": "2023-05-30T15:58:25"
    },
    "payDetails": {
      "atomTxnId": 11000000379843,
      "prodDetails": [
        {
          "prodName": "AIPAY",
          "prodAmount": 100
        }
      ],
      "amount": 100,
      "surchargeAmount": 0,
      "totalAmount": 100,
      "custAccNo": "213232323",
      "clientCode": "1234",
      "txnCurrency": "INR",
      "signature":
"30ff1ac95e9ab010415211def21ca35c81c471d95a8b0dbab437b0bdd33d1822b9cea0d7720396352b26954e
1bf39a1f1cf37ed97909ed7a4fed169110ce357d",
      "txnInitDate": "2023-05-30 15:58:26",
      "txnCompleteDate": "2023-05-30 15:58:41"
    },
    "payModeSpecificData": {
      "subChannel": [
        "UP"
      ],
      "bankDetails": {
        "otsBankId": 2,
        "bankTxnId": "1685442502736",
        "authId": "7401685442502736",
        "otsBankName": "Hdfc Bank"
      }
    }
  }
}

```

```

    }
  },
  "extras": {},
  "custDetails": {
    "billingInfo": {}
  },
  "responseDetails": {
    "statusCode": "OTS0000",
    "message": "SUCCESS",
    "description": "APPROVED OR COMPLETED SUCCESSFULLY"
  }
}

```

**NEFT/RTGS Challan Response Sample format:**

```

{
  "payInstrument": {
    "merchDetails": {
      "merchId": 317159,
      "merchTxnId": "test000123",
      "merchTxnDate": "2023-05-30T16:00:31"
    },
    "payDetails": {
      "atomTxnId": 11000000379847,
      "prodDetails": [
        {
          "prodName": "AIPAY",
          "prodAmount": 100
        }
      ],
      "amount": 100,
      "surchargeAmount": 0,
      "totalAmount": 100,
      "custAccNo": "213232323",
      "clientCode": "1234",
      "txnCurrency": "INR",
      "signature":
"2c1a2e694195129565cc73b1c6c5e13c30daa2c7e7e339a0011ccbde3b850da6ec540da0e1dcf6f6b68be3e1b
9ce37ecbebb3cdf6b04be2cbb2ef850627ee8b9",
      "txnInitDate": "2023-05-30 16:00:32",
      "txnCompleteDate": ""
    },
    "payModeSpecificData": {
      "subChannel": [
        "NR"
      ],
      "bankDetails": {
        "otsBankId": 16,
        "bankTxnId": "ZATOMZ000000379847",

```

```
"authId": "ZATOMZ000000379847"
}
},
"extras": {},
"custDetails": {
  "billingInfo": {}
},
"responseDetails": {
  "statusCode": "OTS0551",
  "message": "PENDING",
  "description": "CHALLAN GENERATED SUCCESSFULLY"
}
}
}
```